The Open Source Audit

A Capstone Project for OSS NGMC Course

Course: Open Source Software
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Software Chosen: Apache HTTP Server

Abstract

Open Source Software (OSS) plays a crucial role in modern computing systems. This project focuses on Apache HTTP Server, one of the most widely used open-source web servers. It includes analysis of its origin, philosophy, licensing model, Linux implementation, and ecosystem. Additionally, shell scripts are developed to demonstrate practical system-level interaction.

1. Introduction

Open-source software allows users to freely use, modify, and distribute software. It promotes innovation and collaboration across communities. Apache HTTP Server is one of the most widely used web servers for hosting websites and applications.

2. Origin and Philosophy

2.1 Origin

Apache HTTP Server was developed in the 1990s by a group of developers aiming to create a stable and reliable web server.

2.2 Philosophy

Apache follows:

- Open collaboration
- Transparency
- Community-driven development
- Freedom of modification

3. License Analysis

Apache uses the Apache License 2.0.

Features:

- Free usage and modification
- Commercial use allowed
- No strict copyleft
- Patent protection

4. Ethical Considerations

- Promotes innovation
- Encourages collaboration
- Risk of misuse
- Security challenges

5. Linux Footprint

Installation:

sudo apt install apache2

Directories:

- /etc/apache2
- /var/log/apache2

Commands:

```
systemctl start apache2
systemctl stop apache2
```

6. Ecosystem

Apache is part of the LAMP Stack:

- Linux
- Apache
- MySQL
- PHP

7. Comparison

Feature	Apache	IIS
Cost	Free	Paid
Source	Open	Closed
Flexibility	High	Low

8. Scripts Overview

- Script 1: System Information
- Script 2: Package Check
- Script 3: Disk Audit
- Script 4: Log Analysis
- Script 5: Manifest Generator

9. Sample Outputs

Figure 1: System Identity Output

System Identity Output

```
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Distro    : Ubuntu 22.04
Kernel    : 5.15.0-105-generic
User      : ubuntu
Uptime    : up 2 hours
Date      : Thu Apr 4 11:32:07 PM IST 2024
License   : GNU GPL (Linux Kernel)
```

Figure 2: Package Inspector Output

Package Inspector Output

```
httpd is installed.

Version    : 2.4.57
Licence    : Apache License 2.0
Summary    : Apache HTTP Server
```

Figure 3: Disk Auditor Output

Disk Auditor Output

```
Directory Audit Report
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/etc => Permissions:  drwxr-rx-r | Size
                            root | 13M

/var/log => Permissions: drwxrxrx-r | Size
                            root | 76M

/home => Permissions: drwxr-rx-r   | Size
                      ubuntu ubuntu | 1.4G
```

Figure 4: Log Analyzer Output

Log Analyzer Output

```
Keyword 'error' found 14 times in
/var/log/syslog

Last 5 matches:

Apr 4 23:29:34 ubuntu apache2 [45318]:
[error] client 192.168.1.100: conecton failed

Apr 4 23:31:05 ubuntu sshd [14324]:
[error] Failed password for root from
203.0.113.5 port 22

Apr 4 23:32:14 ubuntu apache2 [14319]:
[error] request exceeded the limit of
10 internal redrects

Apr 4 23:35:28 ubuntu sshd [4432]:
[error] PAM. authentication error
tor lllegal user admin from 198.51.100.7
```

Figure 5: Manifest Generator Output

Manifest Generator Output

```
Answer the following questions:

Tool you use daily: Bash
Freedom means: ability to modify, share, and improve
  software
What will you build: open source tools

Manifesto saved to manifesto.txt

On 04 April 2024. I believe in open source.
I use Bash reguiarly.
Freedom means ability to modify, share, and improve
  Software
I will build opn source tools and share it with others.
```

10. Results

The scripts demonstrated automation, system monitoring, and log analysis using Linux tools.

11. Conclusion

Apache HTTP Server is a strong example of successful open-source software that enables scalable and reliable web services.

12. References

- Apache Documentation
- Linux Manual Pages
- Open Source Initiative